An AI Pipeline for Image Recognition and Classification

Kristen Thebeau
Worcester Polytechnic Institute
Undergraduate CS & Physics

Sophia Perron
Worcester Polytechnic Institute
Undergraduate Environmental Engineering

Franklin Weiss
Worcester Polytechnic Institute
Undergraduate Physics & Math

Thomas Ostrander
Worcester Polytechnic Institute
Undergraduate Aerospace Engineering

Created on July 10th, 2026 for Professor Marc Bui at L'Université Paris 8 and École Pratique des Hautes Études.

Contents

1 System Requirements

Please note that these instructions are intended for a Windows operating system.

- **CUDA toolkit** - for SAM2 GPU acceleration if using NVIDIA GPU.
- **Python 3.10+**
- **LM Studio** - to locally run Qwen3
- **Conda** - for managing the gsam2 environment

1.1 Python Environment

To be installed in gsam2 conda environment.

- **Torch + Torchvision** - PyTorch, required by SAM2
- **Sam2** - SAM2 package, installed from GitHub repo
- **OpenAI** - Python client used to talk to LM Studio's local server
- **Pillow** - used for image loading and manipulation
- **Numpy** - for array operations
- **Matplotlib** - generated overlay visualizations
- **Tifffile** - for loading .tif files

1.2 Model Files

- SAM2 checkpoint, sam2.1_hiera_large.pt, was used due to acuracy.
  - Small/tiny variants exist, will be quicker but less accurate
  - Installation into checkpoints folder is recommended within the same root directory as pipeline code

    -> root directory
      -> checkpoints folder
        -> sam2.1_hiera_large.pt
      -> stelemotifpipeline2.py
- **SAM2 config** - comes with cloned SAM2 repo
- **Qwen model of choice** - installed and ran within LM Studio

1.3 Code Files

- **stelemotifpipeline2.py** - Custom / edited pipeline combining Qwen3-VL-8B running locally + SAM2 running locally.
- **reference_library.py** - reference crop management and few-shot prompt builder.
- **test_connection.py** - diagnostic only, confirms LM Studio is running and reachable.

Note: stelemotifpipeline2.py and reference_library must be in the same directory.

2 Input and Output

2.1 Input Files

- Input files should in a recognizable location, depicted within pipeline code.
  - Altering stele scans to inverted black and white .tif files is recommended for optimal detection.
    - White on black background may be marginally advantageous for object detection models compared to colored images.
    - Gets reverted for final product.
  - Original colored .jpg file of stele should also be provided for segmentation and final output with bounding boxes and masks.

2.2 Output

- Specify a directory for outputs of the pipeline.
  - Final .jpg of stele with bounding boxes and masks overlaid, separate masks and .json file.

3 Directory Structure

```
AI_Image_detection_Pipeline/
├─ stelemotifpipeline2.py
├─ reference_library.py
├─ test_connection.py
├─ checkpoints/
│  └─ sam2.1_hiera_large.pt
├─ configs/
│  └─ sam2.1/
│     └─ sam2.1_hiera_l.yaml
├─ sam2/ cloned SAM2 repo, installed as package
├─ test_images/
│  ├─ 10925bwi.tif
│  └─ 10925.jpg
└─ output/ generated automatically on first run
   ├─ detection_overlay.png
   ├─ detections.json
   ├─ raw_detections.json
   └─ mask_000_dragon.png
```